

MANUFACTURING TECHNOLOGY

Manufacturing is the transformation of materials into products to meet human needs and wants. Manufacturing technology courses provide students with opportunities to acquire knowledge and skills in the areas of mechanical engineering, robotics and control systems, computer-aided manufacturing, industrial maintenance, precision machining, welding, and sheet metal.

Students will use a broad range of tools and equipment while acquiring engineering, fabrication, and problem-solving skills through the design and fabrication of various projects. Manufacturing courses will also help students understand the effects of manufacturing technology on the environment and society.

Manufacturing courses are designed for students heading to an apprenticeship, to college or university, or directly to the workplace in their pursuit of careers such as tradesperson, technician, technologist, engineer, or any of the skilled entry positions.

The list of approved emphasis areas for manufacturing technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Manufacturing Technology, Grade 12

College Preparation

TMJ4C

This course enables students to further develop knowledge and skills related to machining, welding, print reading, computer numerical control (CNC), robotics, and design. Students will develop proficiency in using mechanical, pneumatic, electronic, and computer control systems in a project-based learning environment and may have opportunities to obtain industry-standard training and certification. Students will expand their awareness of environmental and societal issues and career opportunities in the manufacturing industry.

Prerequisite: Manufacturing Technology, Grade 11, College Preparation

A. MANUFACTURING TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of how the design process is used as a problem-solving strategy to develop solutions to challenges or problems in manufacturing technology;
- A2.** demonstrate an understanding of the processes, tools, and equipment required for making material conversion;
- A3.** demonstrate a working knowledge of various control systems used in manufacturing processes;
- A4.** apply relevant mathematical skills, scientific concepts, and technological literacy and communication skills in the study of manufacturing technology.

SPECIFIC EXPECTATIONS

A1. Design Process

By the end of this course, students will:

- A1.1** demonstrate proficiency in using a design process to solve manufacturing problems;
- A1.2** explain, using specific examples, how the design process has been used in manufacturing to develop new products, or to improve products, to meet human needs or wants (*e.g., development of robotics for use in medical applications has improved surgical procedures and patient outcomes*);
- A1.3** explain why technological concepts (*e.g., aesthetics, control, environmental sustainability/stewardship, ergonomics, fabrication, function, innovation, material, mechanism, power and energy, structure, safety, systems*) are important considerations in the design process (see pp. 7–8);
- A1.4** use technical drawings (*e.g., orthographic, pictorial, assembly drawings and sketches*) as part of the design process to develop solutions to challenges or problems in manufacturing technology;
- A1.5** identify organizations that develop standards in the conception and manufacture of consumer goods (*e.g., Canadian Standards Association [CSA], International Organization for Standardization [ISO]*), and describe how these standards affect the design process.

A2. Material Conversion

By the end of this course, students will:

- A2.1** describe the principles and practical applications of metallurgy (*e.g., hardening, tempering, annealing*);
- A2.2** demonstrate a working knowledge of correct tool geometry and terminology when adapting machine tools to allow for specific manufacturing operations (*e.g., prepare grinding tools with proper relief angles and cutting edges, use proper tool set-up for the application*);
- A2.3** describe in detail the function, purpose, and operation of advanced machine tools and technologies used in manufacturing (*e.g., computerized robot, laser applications, plasma cutting technologies, high-pressure water jet applications*);
- A2.4** explain material conversions such as the separation process (*e.g., converting a material's size and shape by removing excess material*); the addition process (*e.g., combining materials to achieve enhanced qualities, as in alloys*); the process of making changes to contours (*e.g., assembling materials by such means as gluing, mixing, fastening, bonding, welding*); and the process of changing the properties (*e.g., thermal, chemical, mechanical, physical*) of solid materials.

A3. Control Systems

By the end of this course, students will:

- A3.1** demonstrate a working knowledge of how to program computer numerical control (CNC) equipment using computer-aided manufacturing (CAM) software;
- A3.2** demonstrate the correct use of a variety of power and control systems (*e.g., electronic, pneumatic, hydraulic, mechanical systems*);
- A3.3** demonstrate the proper use of various control software programs used in manufacturing processes (*e.g., programmable logic control [PLC], CNC, inventory control software*).

A4. Mathematics, Science, and Technological Literacy

By the end of this course, students will:

- A4.1** apply knowledge of scientific laws (*e.g., Newton's laws of motion, Boyle's law, Ohm's law, Pascal's law*) related to power transmission and mechanical mechanisms (*e.g., hydraulics, pneumatics, gears, pulleys, levers, couplings*) as appropriate during project planning and fabrication;

- A4.2** apply mathematical and technological literacy skills to interpret product specifications (*e.g., volume, area, density/weight, material requirements, scrap factor*) accurately from engineering drawings;

- A4.3** demonstrate effective use of research skills (*e.g., obtaining and evaluating information from technical documents and manufacturing resources*) and technological literacy and communication skills (*e.g., log note writing, record keeping, data recording, memo writing, schedule creation*) to plan and document work on a manufacturing project;

- A4.4** correctly communicate the information needed to plan and prepare for the fabrication process (*e.g., order of operations, availability of tools, parts and equipment required, scheduling requirements, time requirements*), applying relevant technological literacy skills;

- A4.5** estimate the cost of a project, using the appropriate resources (*e.g., material costs, bill of material, estimated labour costs*) and relevant mathematical skills.

B. MANUFACTURING TECHNOLOGY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** demonstrate the ability to interpret and prepare technical drawings and develop process plans;
- B2.** select, prepare, and use appropriate materials when manufacturing a product to meet design specifications;
- B3.** apply quality control standards when using various advanced metrology tools to measure, lay out, and inspect the design and manufacture of products;
- B4.** use appropriate tools and equipment and a variety of manufacturing processes when planning and manufacturing a product to meet specifications.

SPECIFIC EXPECTATIONS

B1. Technical Drawings and Process Plans

By the end of this course, students will:

- B1.1** demonstrate a working knowledge of the stages of the manufacturing process (*e.g., produce an order of operation chart, develop a bill of material for a project, undertake a cost analysis, develop a process plan*);
- B1.2** interpret and create drawings or sketches (*e.g., orthographic projection, assembly, sectional, detail, pictorial*) that include specifications and information, such as symbols and legends, appropriate to the project;
- B1.3** use appropriate resources and reference materials (*e.g., charts, tables, material- and product-related specifications, schematics*) when planning a project;
- B1.4** use a variety of communication techniques (*e.g., multimedia presentation, electronic presentation*) and supporting documentation (*e.g., prints, technical reports, statistical charting*) to present and explain a process design and plan.

B2. Material Selection and Preparation

By the end of this course, students will:

- B2.1** select on the basis of research the appropriate material(s) (*e.g., hot or cold rolled steel, drill rod, ferrous or non-ferrous metals, materials with high or low carbon content, appropriate welding*

electrode and/or gas) for the manufacture of a product (*e.g., low-friction bearings and/or high-traction materials for a robot or robotic component*), based on design criteria;

- B2.2** demonstrate a working knowledge of factors affecting material selection (*e.g., functionality, cost effectiveness, customer expectations, viability, availability*);
- B2.3** use the proper procedures to prepare materials for production (*e.g., measuring, marking, cutting to rough length, grinding, cleaning, deburring*);
- B2.4** test mechanical properties of materials using a tensile tester and hardness tester.

B3. Metrology Tools

By the end of this course, students will:

- B3.1** demonstrate the correct use of common measurement instruments (*e.g., scales, gauge blocks, vernier calipers, inside/outside/depth micrometers, dial indicators, height gauges*) and advanced measuring tools (*e.g., precision level, coordinate measuring machine, laser levels, optical comparators, theodolites, master levels, optic levels, trammel points, opti-line*), using both metric and US customary/British imperial units;
- B3.2** demonstrate the proper use of advanced layout and set-up tools (*e.g., sine bar, gauge blocks, bevel protractor*);

B3.3 use the appropriate measuring tools to precisely measure objects to close tolerance and conduct an ongoing and final inspection process on the attributes of the product with reference to industry metrology standards (*e.g., using statistical process control [SPC] charts and reports*).

B4. Tools, Equipment, and Manufacturing Processes

By the end of this course, students will:

B4.1 demonstrate a working knowledge of the skills required to properly select and safely operate hand tools, machine tools, and equipment (*e.g., wrenches, electric drills, grinders, engine lathe, milling machine*) when manufacturing a product;

B4.2 maintain hand tools, machine tools, and equipment in good and safe working order;

B4.3 perform various manufacturing processes (*e.g., casting, moulding, coating, separating, assembling, cutting*) safely and correctly;

B4.4 demonstrate an understanding of advanced welding and cutting processes (*e.g., gas tungsten arc welding [GTAW], gas metal arc welding [GMAW], shielded metal arc welding [SMAW], plasma cutting, laser cutting, high-pressure water jet cutting*);

B4.5 use various joining processes and/or assembly techniques (*e.g., SMAW, GMAW, GTAW, oxy-acetylene welding, brazing, soldering, tinning, bonding, use of appropriate fasteners*) to complete projects.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of the importance of using sustainable and environmentally friendly manufacturing practices;
- C2.** explain the importance of manufacturing to the economy and society.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** demonstrate the proper selection and disposal of oils, fluids, and materials used in manufacturing;
- C1.2** explain how the three Rs (reduce, reuse, recycle) can minimize the effect the manufacturing industry has on the environment;
- C1.3** explain the advantages and disadvantages of using various renewable and sustainable energy sources (*e.g., solar, hydrogen fuel cell, wind, geothermal, tidal*) in manufacturing;
- C1.4** assess the benefits of using environmentally friendly products and processes in manufacturing (*e.g., long-term cost savings, creation of positive company image by establishing “green” credentials*);
- C1.5** follow environmentally responsible practices during the design and manufacture of a product (*e.g., minimize waste, consider using renewable or recyclable materials, design and manufacture products that last or can be repaired as opposed to throw-away products, use processes that have minimal impact on workers and the local environment*).

C2. Technology and Society

By the end of this course, students will:

- C2.1** explain how the local and/or provincial manufacturing industry has changed over time and describe the effects of these changes on the local and/or provincial economy and society;
- C2.2** explain how global economic shifts are reflected in the local and/or provincial manufacturing industry;
- C2.3** explain how apprenticeships, salaries, standards of living, and the role of skilled trades in society are interrelated;
- C2.4** assess the effects of emerging manufacturing technology industries on culture and society from various perspectives (*e.g., safety, technical, financial, business*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and compliance with health and safety legislation, standards, and practices, including methods to address deficiencies, as they relate to the manufacturing industry;
- D2.** demonstrate an understanding of career opportunities in the manufacturing industry and the training and certification required for these careers.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** demonstrate an understanding of the specific components of legislation and standards related to workplace safety in the manufacturing industry (*e.g., Occupational Health and Safety Act [OHSA], Workplace Hazardous Materials Information System [WHMIS]*) and the specific responsibilities of the relevant oversight/regulatory organizations (*e.g., Workplace Safety and Insurance Board [WSIB], Industrial Accident Prevention Association [IAPA], Construction Safety Association of Ontario [CSAO]*);
- D1.2** describe and assess ways to promote safe and productive work practices in the manufacturing workplace (*e.g., develop and use a safety checklist for work practices, tools, equipment, and operations; develop and conduct safety audits and inspections of the school manufacturing facility; design a plan to address health and safety deficiencies*);
- D1.3** demonstrate good housekeeping practices in the work environment (*e.g., cleaning up spills and leaks, keeping areas clean and clear of obstructions, properly organizing tools and equipment*);
- D1.4** handle materials safely and appropriately in compliance with the information included in the Material Safety Data Sheets (MSDS) from the Workplace Hazardous Materials Information System (WHMIS);

D1.5 use proper ventilation and/or filtration systems to control air quality (*e.g., to minimize the effects of welding fumes, plastic off-gassing, cutting-fluid misting, and heat treating*);

D1.6 explain how ergonomics can affect productivity, product quality, and employee needs and satisfaction, and identify ergonomic considerations related to workshop layout and set-up (*e.g., material handling, ease of movement, lighting, workstation design, tool and equipment organization, speed and efficiency*);

D1.7 use and maintain protective clothing and equipment as required to ensure their own and others' safety in the work environment.

D2. Career Opportunities

By the end of this course, students will:

- D2.1** assess their personal aptitude for career opportunities in the manufacturing industry;
- D2.2** assess postsecondary programs associated with manufacturing and develop a personal pathway leading to a specific career;
- D2.3** research and report on opportunities available locally to obtain certifications and/or training that it would be beneficial to have if they are pursuing a career in manufacturing (*e.g., cardiopulmonary resuscitation [CPR], Standard First Aid, fall protection training*);

D2.4 demonstrate an understanding of and apply the Essential Skills that are important for success in the manufacturing industry, as identified in the Ontario Skills Passport (*e.g., decision making, job task planning and organizing, problem solving*);

D2.5 demonstrate an understanding of and apply the work habits that are important for success in the manufacturing industry, as identified in the Ontario Skills Passport (*e.g., teamwork, working independently, organization*);

D2.6 maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in manufacturing technology (*e.g., Passport to Safety certificate, Ontario Skills Passport Work Plan and Transition Plan, photographs of a project, technical drawings, reports, assignments*), and explain why having a current portfolio is important for career development and advancement.